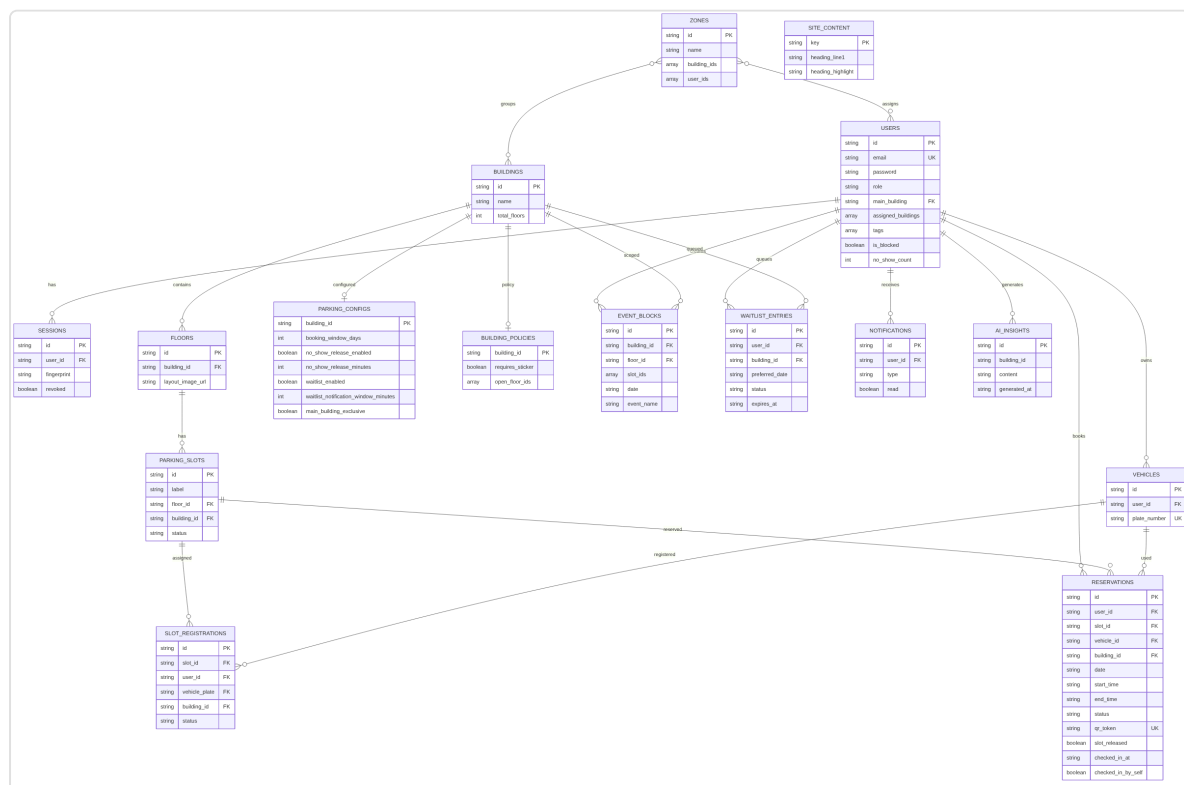


Entity Relationship Diagram

Related: [System Design](#) · Last Updated: May 12, 2026

The ERD below is **database-agnostic**: the same entities exist in both MongoDB (collections) and Couchbase (collections inside the `_default` scope of bucket `db_parking`). The DB Abstraction Layer keeps the application code identical against either backend.

ERD (Mermaid)



Index Summary

The same logical indexes are present on both DB backends; the abstraction layer maps them to the appropriate primitive (MongoDB index spec / Couchbase GSI).

Collection	Indexes
users	<code>id</code> (unique), <code>email</code> (unique)
sessions	<code>id</code> (unique), <code>user_id</code>
reservations	<code>id</code> (unique), <code>user_id</code> , <code>building_id</code> , <code>slot_id</code> , <code>date</code> , <code>qr_token</code> , compound(<code>slot_id+date+status</code>), compound(<code>status+date+slot_released</code>)
buildings	<code>id</code> (unique)
floors	<code>id</code> (unique), <code>building_id</code>
parking_slots	<code>id</code> (unique), <code>floor_id</code>
vehicles	<code>id</code> (unique), <code>user_id</code> , <code>plate_number</code> (unique)
zones	<code>id</code> (unique)
notifications	<code>id</code> (unique), compound(<code>user_id+created_at</code>)
parking_configs	<code>building_id</code> (unique)
building_policies	<code>id</code> (unique), <code>building_id</code> (unique)
slot_registrations	<code>id</code> (unique), compound(<code>slot_id+user_id+status</code>), <code>vehicle_plate</code>
event_blocks	<code>id</code> (unique), compound(<code>building_id+date</code>), compound(<code>floor_id+date</code>)
waitlist_entries	<code>id</code> (unique), compound(<code>building_id+preferred_date+status</code>)
ai_insights	<code>id</code> , <code>generated_at</code>
site_content	<code>key</code> (unique)

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